

AI Hairstyle Intelligence Platform

Professional Product Requirements Document (PRD) Structure

1. Product Overview

1.1 Product Name

HairVerse

1.2 Tagline

Try Before You Cut.

1.3 Product Description

AI-powered realistic hairstyle virtual try-on platform that allows users to preview hairstyles, hair colors, and beard styles using photo upload and live camera AI rendering before getting a haircut.

2. Product Goals

- Help users confidently choose hairstyles before haircut.
 - Provide realistic hairstyle visualization.
 - Deliver personalized hairstyle recommendations.
 - Improve hairstyle decision-making.
 - Create salon-level hairstyle fitting experience digitally.
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3. Target Audience

Primary Users

- Men
- Women
- Teenagers

Secondary Users

- Fashion-conscious users
 - Grooming enthusiasts
 - Hair styling experiment users
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4. Platforms Supported

- Android
 - iOS
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5. User Authentication Module

5.1 Signup Features

- Google Signup
- Email Signup
- Password Creation
- Multi-language onboarding

5.2 Login Features

- Google Login
- Email Login
- Forgot Password
- Remember Me Session
- Secure Authentication

5.3 User Profile Setup

- Name
 - Age
 - Gender
 - Hair Preferences
 - Preferred Hairstyle Types
 - Preferred Hair Length
 - Beard Preference
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6. Onboarding Module

Features

- Welcome Screens
 - App Introduction
 - AI Style Quiz
 - Hairstyle Preference Selection
 - Hair Goals Selection
 - Face Scan Permission
 - Camera Permission
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7. Home Screen Module

Features

- Personalized Hairstyle Feed
 - Trending Hairstyles
 - Recommended Hairstyles
 - Recently Tried Hairstyles
 - Saved Collections
 - Celebrity Inspired Looks
 - Notifications Section
 - Hairstyle Categories
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8. AI Face Analysis Module

AI Analysis Features

- Face Shape Detection
 - Hairline Detection
 - Forehead Analysis
 - Beard Analysis
 - Hair Density Detection
 - Hair Texture Analysis
 - Hair Shine Detection
 - Hair Health Analysis
 - Gray Hair Detection
 - Age-Based Suitability Detection
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9. Hairstyle Recommendation Engine

Features

- Personalized Hairstyle Recommendations
 - Face Shape Matching
 - Hair Texture Matching
 - Beard Compatibility Matching
 - Age-Based Recommendations
 - Hair Density-Based Recommendations
 - Trending Hairstyle Recommendations
 - Celebrity Hairstyle Matching
 - Lifestyle-Based Recommendations
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10. Virtual Hairstyle Try-On Module

Features

- Upload Photo Try-On
 - Live Camera Try-On
 - Realistic Hairstyle Rendering
 - Hair Color Try-On
 - Beard Style Try-On
 - Multiple Hairstyle Comparison
 - Before vs After Comparison
 - Multi-Angle Preview
 - Side Angle Support
 - HD Hairstyle Rendering
 - Accurate Face Preservation
 - Real-Time Hairstyle Switching
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11. Hairstyle Categories Module

Categories

- Fade Hairstyles
- Buzz Cuts
- Curly Hairstyles
- Korean Hairstyles
- Professional Hairstyles
- Long Hairstyles

- Wolf Cut
 - Layer Cut
 - Wedding Hairstyles
 - Trending Hairstyles
 - Celebrity Hairstyles
 - Beard Styles
 - Hair Color Styles
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12. Hairstyle Search & Filter Module

Features

- Search Hairstyles
 - Filter by Face Shape
 - Filter by Hair Type
 - Filter by Hair Length
 - Filter by Hair Texture
 - Filter by Trend
 - Filter by Gender
 - Filter by Celebrity
 - Filter by Beard Compatibility
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13. Hair Insights Module

Features

- Hair Length Indicator
 - Hair Health Suggestions
 - Hair Growth Suggestions
 - Hair Damage Insights
 - Dryness Detection
 - Hair Thickness Insights
 - Hair Shine Insights
 - Gray Hair Insights
 - Haircare Recommendation Tips
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14. Saved Collections Module

Features

- Save Favorite Hairstyles
 - Create Hairstyle Collections
 - Recently Tried History
 - Saved Hair Colors
 - Saved Beard Styles
 - Personalized Style Library
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15. Notification System

Features

- New Hairstyle Recommendations
 - Trending Hairstyle Alerts
 - Personalized Notifications
 - Hairstyle Suggestions Based on Preferences
 - AI Recommendation Alerts
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16. Settings Module

Features

- Language Selection
 - Dark Mode
 - Privacy Settings
 - Notification Controls
 - Account Management
 - Delete Account
 - Auto Delete Uploaded Selfies
 - Security Settings
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17. Privacy & Security Module

Features

- Secure Authentication

- Encrypted User Data
 - Privacy Mode
 - Auto Delete Selfies
 - Secure Cloud Storage
 - User Consent Management
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18. Multi-Language Support Module

Features

- English
 - Tamil
 - Hindi
 - Additional Global Languages
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19. AI System Features

Core AI Features

- Face Detection
 - Face Landmark Tracking
 - Hair Segmentation
 - Hairstyle Rendering
 - Hair Color Rendering
 - Beard Rendering
 - Recommendation Engine
 - Personalized AI Suggestions
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20. User Experience Features

UX Features

- Futuristic UI Design
 - Smooth Animations
 - Fast Loading
 - Realistic Rendering Experience
 - Interactive Camera Experience
 - Minimal Navigation
 - Mobile-Optimized UI
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21. Monetization Model

Freemium Features

- Limited Free Hairstyle Try-Ons
- Basic Recommendations
- Limited HD Exports

Premium Features

- Unlimited Try-Ons
 - Advanced Realistic Rendering
 - Premium Hairstyle Collections
 - Advanced AI Recommendations
 - Unlimited HD Exports
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22. Technical Stack

Frontend

- React Native

Backend

- Python
- FastAPI

Database

- Firebase Firestore

Authentication

- Firebase Authentication

Storage

- Firebase Storage

Computer Vision

- MediaPipe
- OpenCV

AI Rendering

- Stable Diffusion
 - ControlNet
 - IPAdapter
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23. Future Expansion Scope

Future Features

- Clothing Try-On
 - Makeup Try-On
 - Eyewear Try-On
 - AI Wardrobe System
 - Full Appearance Intelligence Platform
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24. Core Product Differentiators

- Realistic Hairstyle Rendering
 - Accurate Face Preservation
 - Personalized AI Recommendations
 - Multi-Angle Hairstyle Visualization
 - Hair Health Intelligence
 - AI Hairstyle Matching
 - Futuristic User Experience
 - Real-Time Virtual Hair Experience
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25. Main Problem Solved

Current hairstyle apps:

- look unrealistic
- change user face
- lack personalization
- provide inaccurate hairstyle fitting

HairVerse solves this by providing:

- realistic hairstyle fitting
- accurate face preservation

- personalized hairstyle intelligence
- AI-powered hairstyle recommendations before haircut.